Spearman Rank Correlation Test

2026-04-17

Introduction

Spearman's ρ (rho) is the Pearson correlation of the ranks of two variables. It detects monotonic associations that need not be linear, and is robust to outliers and heavy tails. It is the default correlation measure for ordinal data or strongly skewed continuous data.

Prerequisites

Pearson correlation, ranks.

Theory

Replace each X_i by its rank R_i, and each Y_i by its rank S_i. Spearman's ρ is

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}, \qquad d_i = R_i - S_i,$$

when there are no ties. With ties, the general form

$$\rho = \frac{\sum (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum (R_i - \bar{R})^2 \sum (S_i - \bar{S})^2}}$$

applies.

Under $H_0 : \rho = 0$, an asymptotic normal approximation holds; for small n exact or permutation distributions are used. Confidence intervals for Spearman's ρ can be obtained by Fisher's z applied to the ranks.

Assumptions

- Ordinal or continuous data; the ranks must be well-defined.
- Monotonic association (linear or non-linear).
- Independent pairs.

R Implementation

```r
set.seed(2026)

n <- 50
x <- rgamma(n, shape = 2, rate = 0.5)      # skewed
y <- x^2 + rnorm(n, sd = 4)                # non-linear monotonic

# Pearson (fails for non-linear)
cor.test(x, y, method = "pearson")
```

```r
# Spearman
cor.test(x, y, method = "spearman", exact = FALSE)

# Bootstrap 95% CI for Spearman's rho
library(boot)
rho_boot <- boot(data = data.frame(x, y),
                 statistic = function(d, i) cor(d$x[i], d$y[i], method = "spearman"),
                 R = 2000)
boot.ci(rho_boot, type = "perc")
```

Output & Results

```
        Pearson's product-moment correlation
t = 11.45, df = 48, p < 0.001
cor: 0.855

        Spearman's rank correlation rho
S = 100, p < 0.001
rho: 0.995

Bootstrap 95% CI (percentile): (0.988, 0.999)
```

Spearman recovers the strong monotonic relationship ($\rho \approx 1$) that Pearson underestimates because of the non-linearity.

Interpretation

"There was a strong monotonic association between x and y (Spearman's $\rho = 0.99$, bootstrap 95 % CI 0.99 to 1.00; $p < 0.001$)."

Practical Tips

- Use Spearman for ordinal data, heavy-tailed or skewed continuous data, or suspected non-linear monotonic relationships.
- For very small n, use exact methods (`cor.test(..., exact = TRUE)`; only valid without ties).
- With ties, the normal approximation is used; the warning is expected and safe.
- Kendall's τ is an alternative for small samples or many ties.
- Spearman captures monotonic, not linear, associations; for U-shaped relationships it can still miss dependence.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation

- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests